

# Thinking Beyond Images

Using **Language** to Improve **Image Classification**

---

Javon Hickmon, Sarah Pratt, Prof. Ali Farhadi

**W** RAIVN Lab

# What is Multimodal Machine Learning?

# Machine Learning

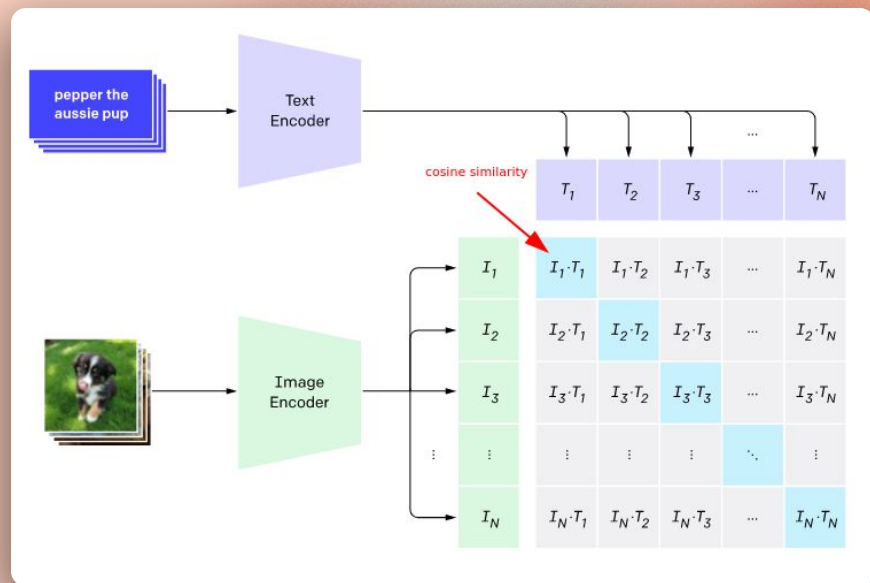
- The use and development of computer systems that can learn and adapt by using algorithms and statistical models to analyze underlying patterns in data.
- Machine Learning (ML) is used in technology all around us



**ChatGPT**

# Multimodal Learning

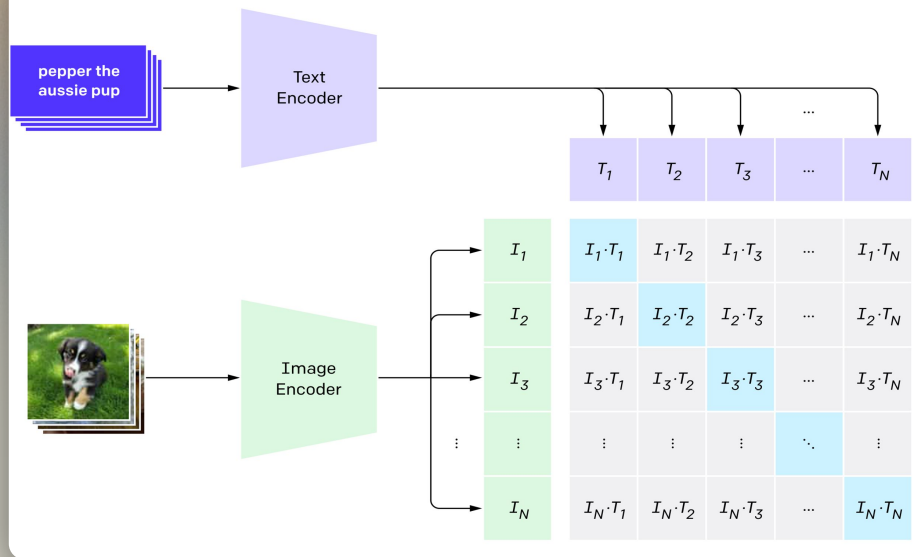
- Is a form of Machine Learning that aims to train models that learn patterns between multiple forms (or modalities) of data
- This could be images to text, audio to text, numerical data to images, etc.
- A common neural network architecture for Multimodal Learning is called CLIP



The CLIP (Contrastive Language-Image Pretraining) Architecture

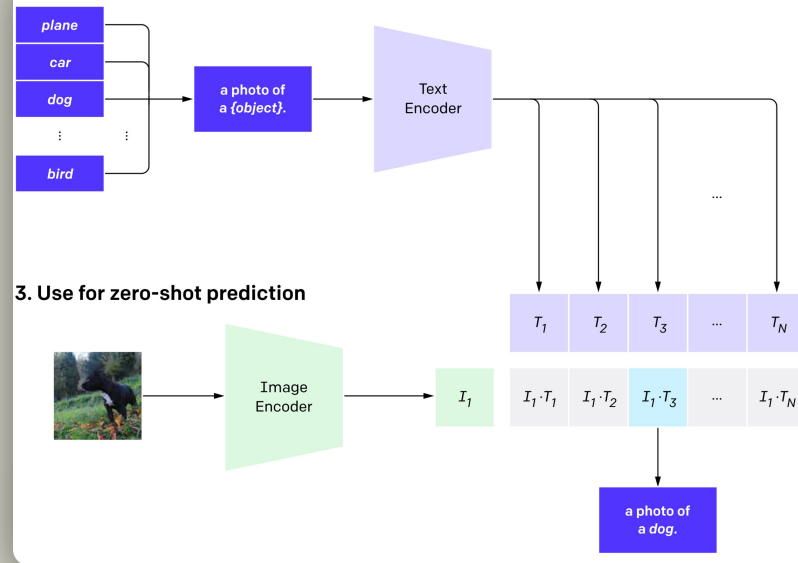
# Training

## 1. Contrastive pre-training



# Inference

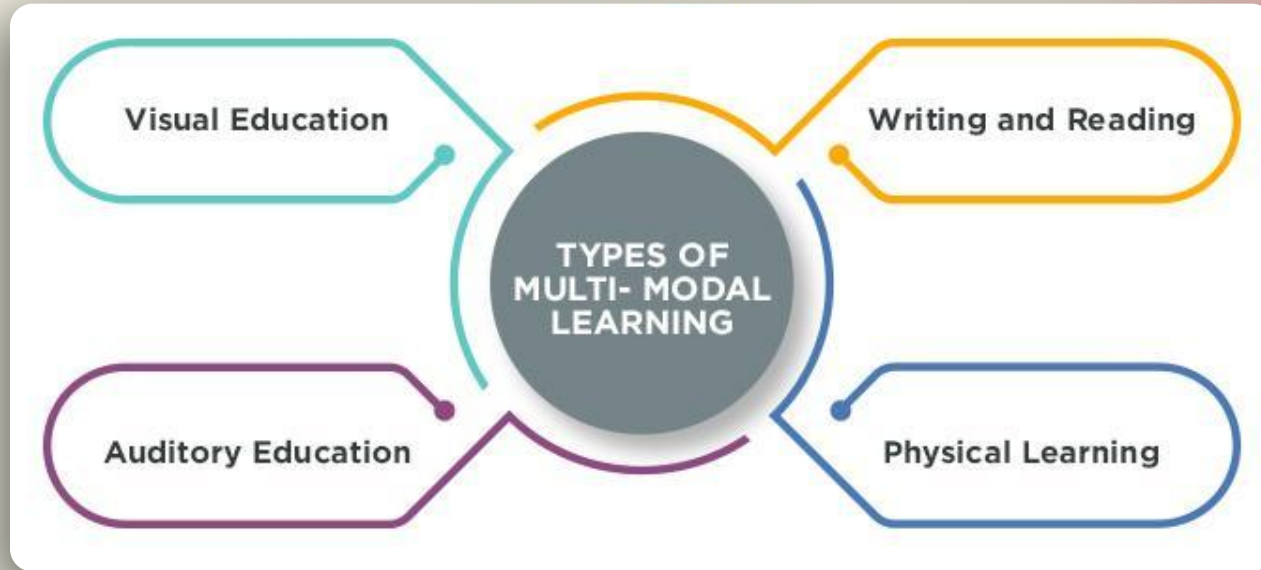
## 2. Create dataset classifier from label text



## 3. Use for zero-shot prediction

# Why is Multimodal Learning Important?

# Multimodal Learning is more than just ML



“The study found that 1st year students were using unimodal learning approach as year of study increased; the students were shifted toward multimodal of learning”  
(Bokhari et al., 2019)

# The Problem

What's the best way to **transfer knowledge** from  
one modality to another?

# Research Methods

## OpenFlamingo

Our Multimodal Model

## Chain-of-Thought

Our prompting technique

## CuPL

Specialized image  
descriptions

# Chain-of-Thought

- A series of intermediate reasoning steps
- By structuring our prompts in specific way, we can have Large Language Models (LLMs) generate their own Chain-of-Thought
- Has been shown chain-of-thought prompting significantly improves the ability of LLMs to perform complex reasoning

## Chain-of-Thought Prompting

### Model Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls.  $5 + 6 = 11$ . The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

### Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had  $23 - 20 = 3$ . They bought 6 more apples, so they have  $3 + 6 = 9$ . The answer is 9. ✓

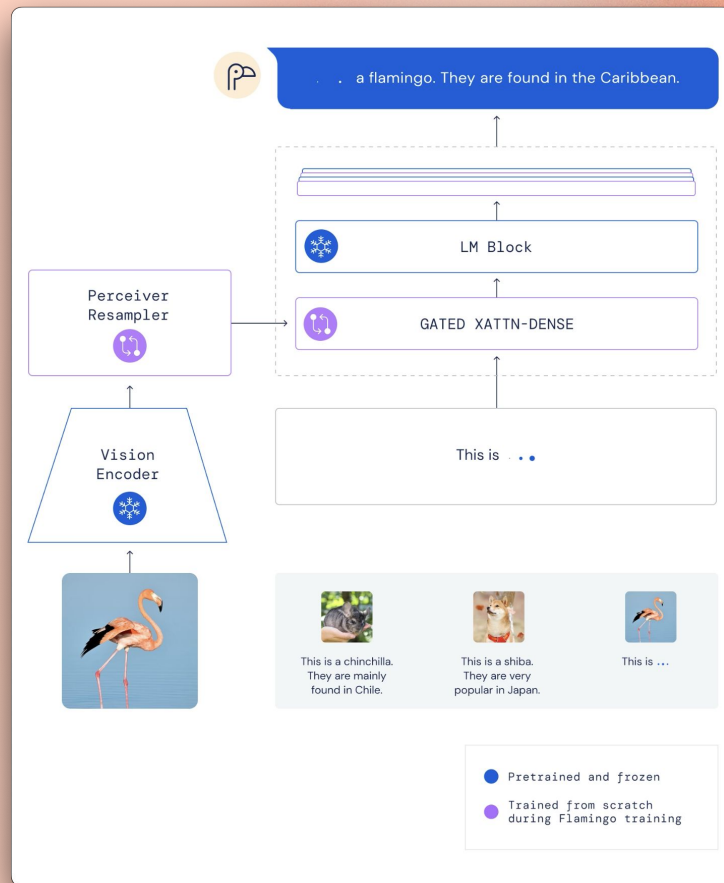
The Chain-of-Thought Prompting Structure

# Lesson Learned

Chain-of-thought prompting can be used to elicit **multistep reasoning** in Large Language Models

# OpenFlamingo

- Flamingo is a Multimodal Vision Language Model developed by Google DeepMind
- Essentially functions like a chatbot that can also accept images
- Recent efforts have been able to create an open-source version of this architecture called OpenFlamingo



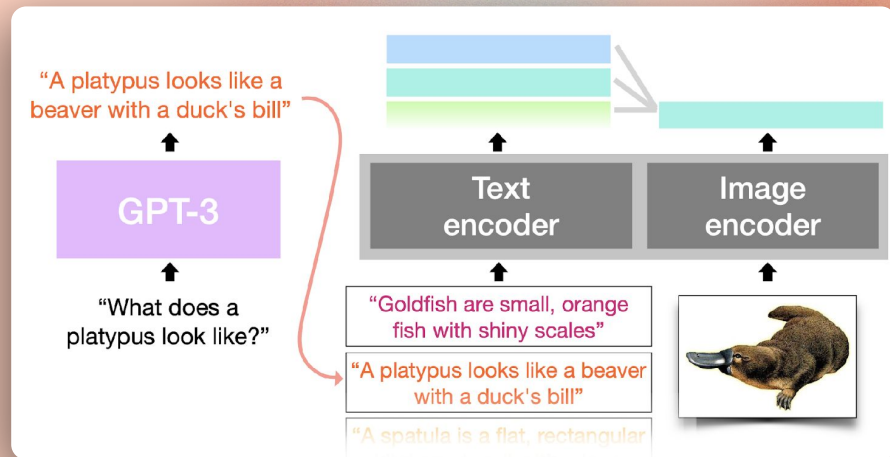
The Flamingo Architecture

# Lesson Learned

We can connect the **semantic knowledge** learned from text and images during inference by using a Multimodal Model

# CuPL

- Customized Prompts via Language models
- A technique developed to generate highly descriptive prompts for the task of zero-shot image classification
- Has been shown to improve accuracy on a range of zero-shot image classification benchmarks, including over one percentage point gain on ImageNet



The CuPL (Customized Prompts via Language models) Architecture

# Lesson Learned

**Knowledge** learned from one modality can directly be **transferred** and improve another

# Our Question

Do the benefits from **Chain-of-Thought** carry over to the **Multimodal** domain?

# Our Approach



What is in this picture?



There is a feline with short brown and white fur and pointy ears.



Therefore, this is a European shorthair.



What is in this picture?



There is a canine with short golden and white fur and pointy ears.



Therefore, this is a Welsh Corgi.



What is in this picture?



There is a duck-billed mammal with brown fur.



Therefore, this is a platypus.

# Preliminary Results

“What is in this photo?”

**Standard Approach:** A photo of a wine bottle

**Our Approach:** There is a small, hand-held device that is used to open bottles, so there is a corkscrew in this image



# Next Steps

## Analyze

common mistakes made by the model, and adjust the prompting structure to reduce inaccuracies

## Quantify

the accuracy on a range of image classification benchmarks

## Observe

whether or not we can still achieve state-of-the-art multimodal performance

# Thank You!



Sarah Pratt



Prof. Ali Farhadi

And a special thank you to the UW McNair Program

# References

- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., ... & Sutskever, I. (2021, July). Learning transferable visual models from natural language supervision. In International conference on machine learning (pp. 8748-8763). PMLR.
- Mathur, Vrinda. "Multi-Modal Learning: Types, Benefits & Examples." Analytics Steps, 29 Aug. 2022, [www.analyticssteps.com/blogs/multi-modal-learning-types-benefits-examples](http://www.analyticssteps.com/blogs/multi-modal-learning-types-benefits-examples).
- Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., ... & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35, 24824-24837.
- Alayrac, Jean-Baptiste, et al. "Tackling Multiple Tasks with a Single Visual Language Model." Deepmind, 28 Apr. 2022, [www.deepmind.com/blog/tackling-multiple-tasks-with-a-single-visual-language-model](http://www.deepmind.com/blog/tackling-multiple-tasks-with-a-single-visual-language-model).
- Pratt, S., Liu, R., & Farhadi, A. (2022). What does a platypus look like? generating customized prompts for zero-shot image classification. *arXiv preprint arXiv:2209.03320*.